Electrode-electrolyte interface — Solution

X23 (2023) — English translation

A1 — 0.02 pts

Measure the inner diameter of the syringe D_0 and the distance L_0 between marks 0 mL and 20 mL.

Answer: $L_0 = 65\,\text{mm}$

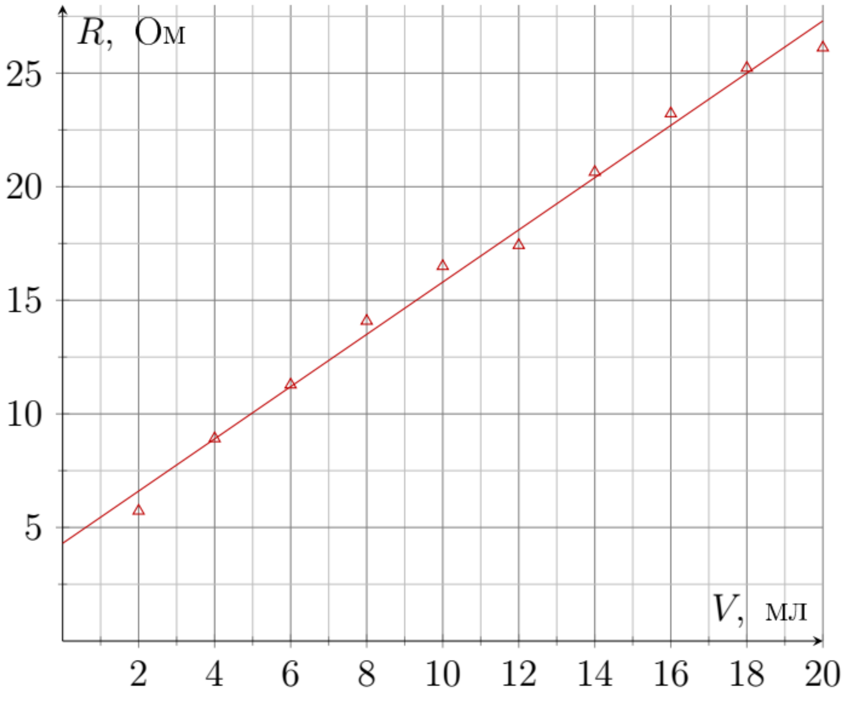
A2 — 0.40 pts

Make resistance measurements R corresponding to the positions of the syringe $V = 2, 4, \ldots, 20$ mL.

A3 — 0.40 pts

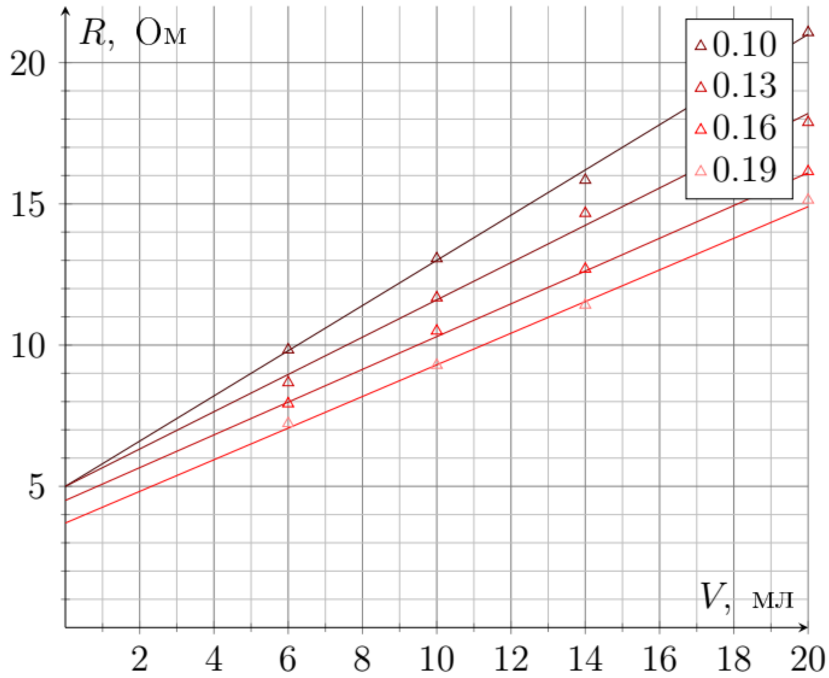
Plot R versus V. Draw a smoothing straight line $R = R_1 + \alpha_1 V$. Calculate the values of parameters R_1 and α_1.

Answer: $R_1 = 4.3\,\Omega$



B1 — 0.80 pts

For each of the prepared solutions, carry out a series of 4-ex measurements R from V.



B2 — 0.12 pts

*Using least squares method for each solution, calculate the parameters R_1 and α_1 by analogy with point **A3**. Present the results in the form of a table.*

c	0.07	0.1	0.13	0.16	0.19
α_1, $\frac{\Omega}{cm^3}$	1.15	0.80	0.66	0.58	0.56
R_1, Ω	4.3	5.0	4.9	4.5	3.7

B3 — 0.06 pts

An external constant field E is created in the medium. What is the steady-state velocity of the particle v_{at}? Write the answer using E, k, q.

Let's write down Newton's II law:

$$Eq = kv_y$$

Answer:

$$v_y = \frac{Eq}{k}$$

B4 — 0.15 pts

What is the steady-state current I if a potential difference U is applied to the electrodes? Neglect any surface effects (eg contact potential difference and contact resistance). Write your answer using $n, S, L, e, k_{Na^+}, k_{Cl^-}$ and U.

By definition $I = \frac{dq}{dt}$, in this case, a flow of particles with an area S and a charge q moving at a constant speed v forms the following current:

$$I = \frac{dq}{dt} = nvSq \tag{1}$$

Then the current in our case consists of two ion flows. The ion concentrations are equal from the condition of electrical neutrality:

$$I = nSe\frac{Ee}{k_{Na^+}} + nSe\frac{Ee}{k_{Cl^-}}.$$

The field E is expressed through the potential difference as U/L.

Answer:

$$I = \frac{nSe^2U}{L}\left(\frac{1}{k_{\mathrm{Na}^+}} + \frac{1}{k_{\mathrm{Cl}^-}}\right)$$

B5 — 0.65 pts

Let a potential difference $U(t) = U\cos\omega t$ be applied to the bases, how will the current $I(t)$ depend on time? Again, ignore any surface effects. Write your answer using $n, S, L, e, \omega, m_{\mathrm{Na}^+}, m_{\mathrm{Cl}^-}, k_{\mathrm{Na}^+}, k_{\mathrm{Cl}^-}, U$ and t.

We will solve the problem using the method of complex amplitudes:

$$U\cos\omega t = \mathfrak{Re}\, Ue^{i\omega t}.$$

Then the complex electric field E inside the medium is:

$$\tilde{E} = \frac{U}{L}e^{i\omega t}.$$

We write Newton's II law for a particle in an alternating field $Ee^{i\omega t}$:

$$m\dot{v} + kv = Eqe^{i\omega t}.$$

Our own solutions are exponentially decaying, so we are only interested in a particular solution of the form $v = \tilde{B}e^{i\omega t}$:

$$i\omega m\tilde{B} + k\tilde{B} = Eq \quad \Rightarrow \quad \tilde{B} = \frac{Eq}{i\omega m + k} = \frac{Eq}{k}\frac{1}{1 + i\omega\frac{m}{k}} = \frac{Eq}{k}\frac{1}{\sqrt{1 + \frac{\omega^2 m^2}{k^2}}}e^{-i\arctan\frac{\omega m}{k}}.$$

Complex Let's recalculate the current $\tilde{I}$ taking into account equation (1):

$$\tilde{I} = \frac{nSe^2U}{Lk_{\mathrm{Na}^+}}\frac{1}{\sqrt{1 + \frac{\omega^2 m^2_{\mathrm{Na}^+}}{k^2_{\mathrm{Na}^+}}}}e^{-i\arctan\frac{\omega m_{\mathrm{Na}^+}}{k_{\mathrm{Na}^+}} + i\omega t} + \frac{nSe^2U}{Lk_{\mathrm{Cl}^-}}\frac{1}{\sqrt{1 + \frac{\omega^2 m^2_{\mathrm{Cl}^-}}{k^2_{\mathrm{Cl}^-}}}}e^{-i\arctan\frac{\omega m_{\mathrm{Cl}^-}}{k_{\mathrm{Cl}^-}} + i\omega t}$$

Let's move from the complex current by taking the real part.

Answer:

$$I = \frac{nSe^2U}{Lk_{\mathrm{Na}^+}}\frac{1}{\sqrt{1 + \frac{\omega^2 m^2_{\mathrm{Na}^+}}{k^2_{\mathrm{Na}^+}}}}\cos\left(-\arctan\frac{\omega m_{\mathrm{Na}^+}}{k_{\mathrm{Na}^+}} + \omega t\right) + \frac{nSe^2U}{Lk_{\mathrm{Cl}^-}}\frac{1}{\sqrt{1 + \frac{\omega^2 m^2_{\mathrm{Cl}^-}}{k^2_{\mathrm{Cl}^-}}}}\cos\left(-\arctan\frac{\omega m_{\mathrm{Cl}^-}}{k_{\mathrm{Cl}^-}} + \omega t\right)$$

B6 — 0.25 pts

Now let's take into account surface effects in the form of a resistor R_m, which is connected in series to the solution. Write down the formula for the impedance $\tilde{Z}(\omega)$ of a system consisting of a solution NaCl *and a vessel. Write your answer using $n, S, L, e, m_{\mathrm{Na}^+}, m_{\mathrm{Cl}^-}, \omega, k_{\mathrm{Na}^+}, k_{\mathrm{Cl}^-}$ and R_m.*

The impedance of one type of ion is obviously expressed from the already obtained equations

$$\tilde{Z} = \frac{Lk}{nSq^2}\left(1 + i\omega\frac{m}{k}\right),$$

, while the currents add up at the same voltages, which means that impedances Na^+ and Cl^- are "connected" in parallel, and there is also a resistor R_m in series with them.

Answer:

$$\tilde{Z}(\omega) = R_\mathrm{m} + \frac{\frac{k_{\mathrm{Na}^+}L}{nSe^2}\left(1 + \frac{i\omega m_{\mathrm{Na}^+}}{k_{\mathrm{Na}^+}}\right)\cdot\frac{k_{\mathrm{Cl}^-}L}{nSe^2}\left(1 + \frac{i\omega m_{\mathrm{Cl}^-}}{k_{\mathrm{Cl}^-}}\right)}{\frac{k_{\mathrm{Na}^+}L}{nSe^2}\left(1 + \frac{i\omega m_{\mathrm{Na}^+}}{k_{\mathrm{Na}^+}}\right) + \frac{k_{\mathrm{Cl}^-}L}{nSe^2}\left(1 + \frac{i\omega m_{\mathrm{Cl}^-}}{k_{\mathrm{Cl}^-}}\right)}$$

B7 — 0.10 pts

What is the theoretical system impedance $\tilde{Z}$ if the phase difference between the voltages is negligible? Express your answer in terms of $n, S, L, e, k_{\mathrm{Na}^+}, k_{\mathrm{Cl}^-}$ and R_m.

If the phase difference is negligible, then the complex parts can be neglected and the behavior of the system corresponds to the behavior at a constant voltage.

Answer:

$$\tilde{Z} = R_{\mathrm{m}} + \frac{L}{nSe^2}\frac{k_{\mathrm{Na}^+}\cdot k_{\mathrm{Cl}^-}}{k_{\mathrm{Na}^+}+k_{\mathrm{Cl}^-}}$$

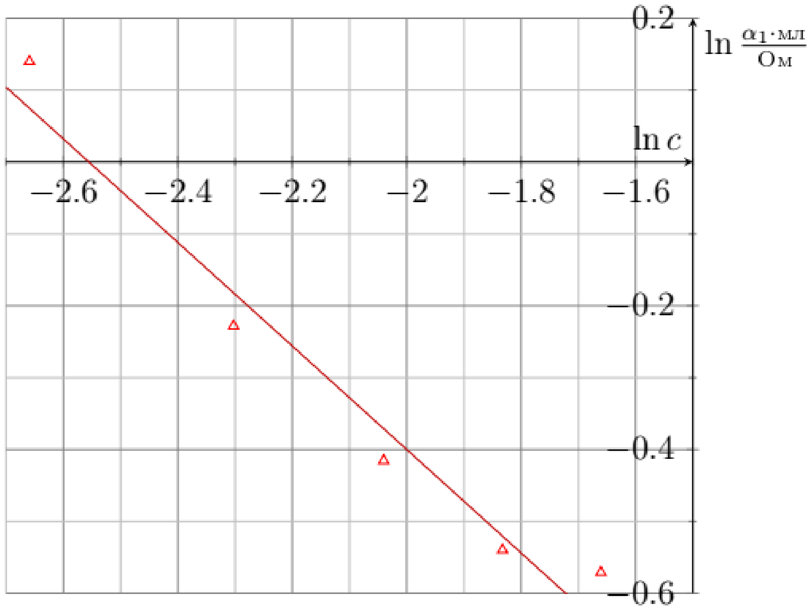
B8 — 0.40 pts

Using the linearized graph of α_1 against c obtain the value α.

Answer:

$$\alpha_1 = \frac{dZ}{dL}\cdot\frac{1}{S}\propto\frac{k}{n}\propto n^{\alpha-1}$$

From graph $\alpha_1 = 0.3$.



B9 — 0.40 pts

Obtain the theoretical dependence $|\tilde{Z}|^2(\omega)$ in the approximation $\omega m_{\mathrm{Na}^+}\gg k_{\mathrm{Na}^+}$, $\omega m_{\mathrm{Cl}^-}\gg k_{\mathrm{Cl}^-}$. Write your answer using $n, S, L, e, m_{\mathrm{Na}^+}, m_{\mathrm{Cl}^-}, \omega, k_{\mathrm{Na}^+}, k_{\mathrm{Cl}^-}$ and R_m.

Let's simplify the formula for impedance from B6:

$$\tilde{Z} = R_{\mathrm{m}} + \frac{L}{nSe^2}\frac{(k_{\mathrm{Na}^+}+i\omega m_{\mathrm{Na}^+})(k_{\mathrm{Cl}^-}+i\omega m_{\mathrm{Cl}^-})}{(k_{\mathrm{Na}^+}+k_{\mathrm{Cl}^-})+i\omega(m_{\mathrm{Na}^+}+m_{\mathrm{Cl}^-})}$$

The first approximation occurs at the multiplication stage:

$$\tilde{Z} = R_{\mathrm{m}} + \frac{L}{nSe^2}\frac{-\omega^2 m_{\mathrm{Na}^+}m_{\mathrm{Cl}^-}+i\omega(m_{\mathrm{Na}^+}k_{\mathrm{Cl}^-}+m_{\mathrm{Cl}^-}k_{\mathrm{Na}^+})}{(k_{\mathrm{Na}^+}+k_{\mathrm{Cl}^-})+i\omega(m_{\mathrm{Na}^+}+m_{\mathrm{Cl}^-})}$$

Multiply the numerator and denominator by the complex conjugate of the denominator and neglect again:

$$\tilde{Z} = R_{\mathrm{m}} + \frac{L}{nSe^2}\frac{-\omega^2 m_{\mathrm{Na}^+}m_{\mathrm{Cl}^-}(k_{\mathrm{Na}^+}+k_{\mathrm{Cl}^-})+i\omega^3 m_{\mathrm{Na}^+}m_{\mathrm{Cl}^-}(m_{\mathrm{Na}^+}+m_{\mathrm{Cl}^-})+\omega^2(m_{\mathrm{Na}^+}k_{\mathrm{Cl}^-}+m_{\mathrm{Cl}^-}k_{\mathrm{Na}^+})(m_{\mathrm{Na}^+}+m_{\mathrm{C}}}{\omega^2(m_{\mathrm{Na}^+}+m_{\mathrm{Cl}^-})^2}$$

Answer:

$$|\tilde{Z}|^2 = \left(R_\mathrm{m} + \frac{L}{nSe^2}\frac{m_{\mathrm{Na}^+}^2 k_{\mathrm{Cl}^-} + m_{\mathrm{Cl}^+}^2 k_{\mathrm{Na}^+}}{(m_{\mathrm{Na}^+} + m_{\mathrm{Cl}^-})^2}\right)^2 + \omega^2\left(\frac{L}{nSe^2}\frac{m_{\mathrm{Na}^+} m_{\mathrm{Cl}^-}}{m_{\mathrm{Na}^+} + m_{\mathrm{Cl}^-}}\right)^2$$

B10 **0.35 pts**

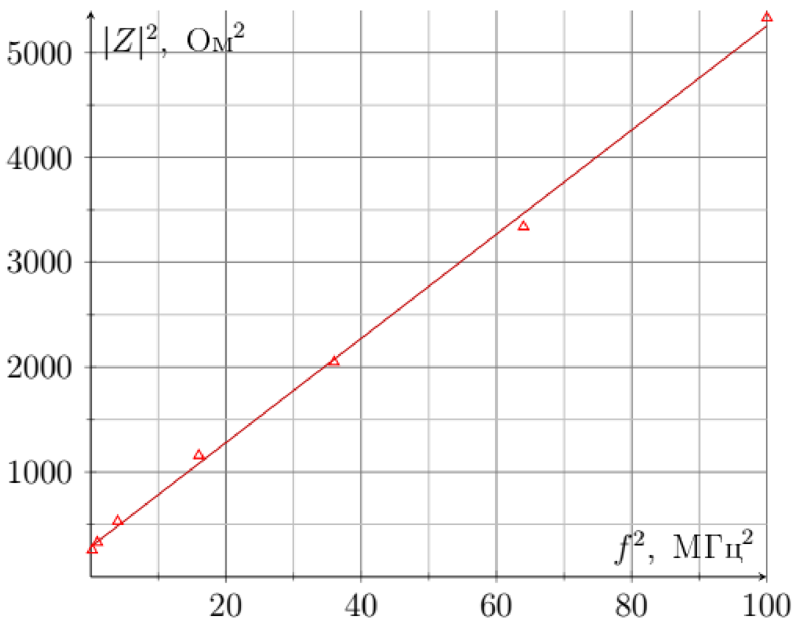
For a solution with salt concentration $c \approx 0.20$, measure the impedance modulus $|Z|$ versus frequency f. The series must consist of at least 7 different f in the range 500 kHz $\le f \le 10$ MHz.

B11 **0.50 pts**

Plot a linearized graph of $|Z|$ versus f. From the graph, determine the experimental value $\mu_{eff} = \frac{\mu_{\mathrm{Na}}\cdot\mu_{\mathrm{Cl}}}{\mu_{\mathrm{Na}}+\mu_{\mathrm{Cl}}}$.

According to the last theoretical point, the graph of $|\tilde{Z}|^2$ versus f^2 is linear.

Answer: From the graph, the slope coefficient is $61 \cdot 10^{-12}\ \frac{\Omega^2}{\mathrm{Hz}^2}$, and from theory it is $\left(\frac{2\pi L m_{\mathrm{eff}}}{nSe^2}\right)^2$. $\mu_{eff} = 170 \cdot 10^5\ \frac{\mathrm{kg}}{\mathrm{mol}}$



B12 **0.15 pts**

Determine N based on your experimental data.

Let's write the formula based on the model described in the condition:

$$\mu_{\mathrm{eff}} = \frac{(\mu_{\mathrm{Na}} + N\mu_{\mathrm{H_2O}})\cdot(\mu_{\mathrm{Cl}} + N\mu_{\mathrm{H_2O}})}{\mu_{\mathrm{Na}} + N\mu_{\mathrm{H_2O}} + \mu_{\mathrm{Cl}} + N\mu_{\mathrm{H_2O}}}$$

Let's transform it to a quadratic equation for N:

$$N^2 + N\left(\frac{\mu_{\mathrm{Na}} + \mu_{\mathrm{Cl}} - 2\mu_{\mathrm{eff}}}{\mu_{\mathrm{H_2O}}}\right) + \frac{\mu_{\mathrm{Na}}\mu_{\mathrm{Cl}} - \mu_{\mathrm{eff}}(\mu_{\mathrm{Na}} + \mu_{\mathrm{Cl}})}{\mu_{\mathrm{H_2O}}^2} = 0$$

Due to the fact that μ_{eff} is very larger, in fact, $N = \frac{2\mu_{\mathrm{eff}}}{\mu_{\mathrm{H_2O}}} = 1.8 \cdot 10^7$. It can be concluded that the measured inductance is in no way related to the theoretically considered effect. Meanwhile, the inductance of a turn with a current radius of 10 cm can be estimated as 10^{-6} H, which is of the same order as the root of the slope coefficient.

Answer:

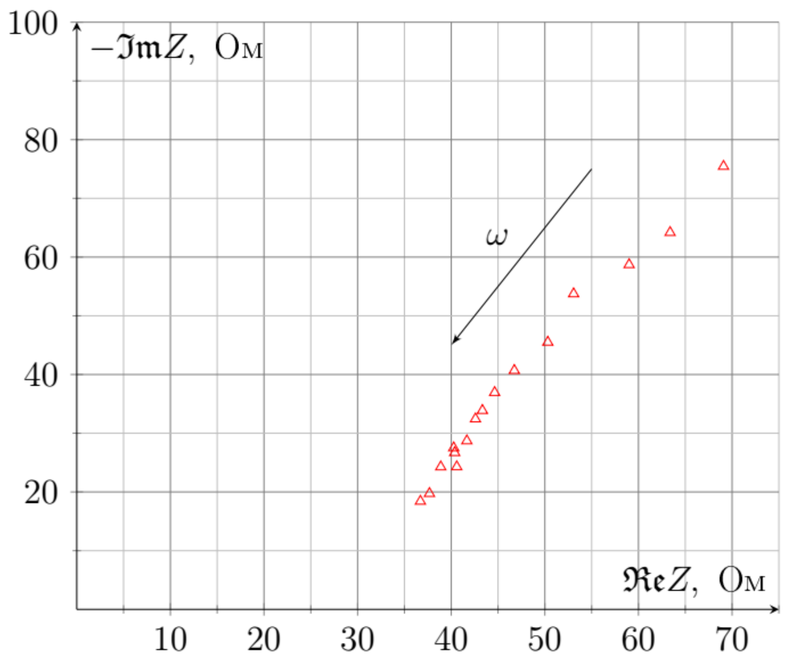
$$N = 1.8 \cdot 10^{7}$$

C1 1.50 pts

In the frequency range 5 Hz $\le f \le 1$ kHz, measure the syringe impedance modulus $|\tilde{Z}|$ and the time Δt by which the current peak precedes the voltage peak. Take at least 15 measurements.

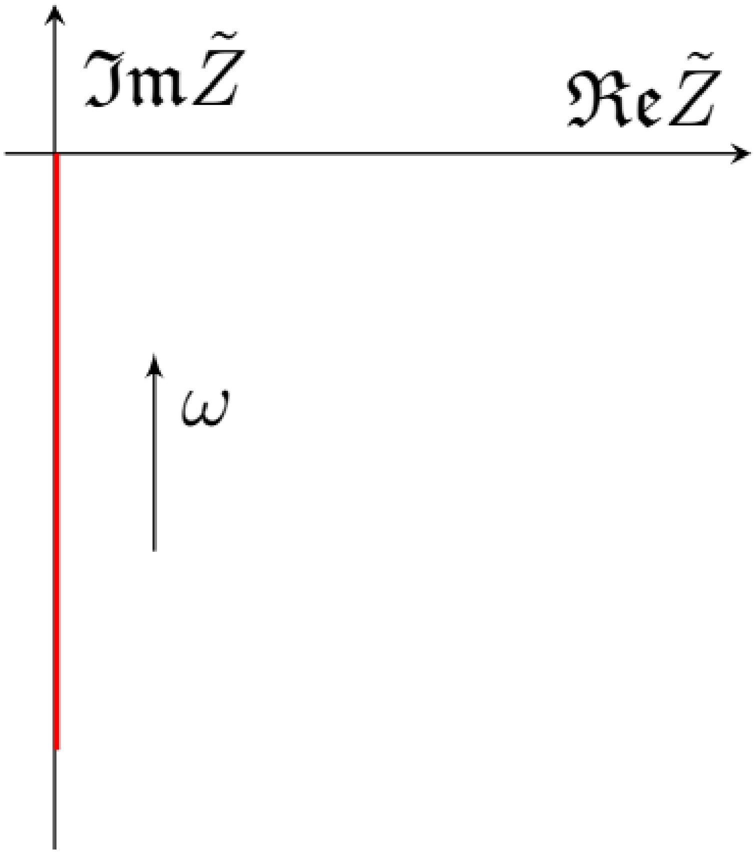
C2 0.60 pts

Based on your data, plot a Nyquist plot of $\mathfrak{Im}\tilde{Z}_m$ from $\mathfrak{Re}\tilde{Z}_m$ for surface effects. Indicate on the diagram in the form of an arrow the direction of transition from point to point, corresponding to the increase ω.



C3 0.10 pts

Qualitatively draw the Nyquist diagram for a parallel-plate capacitor.



C4 — 0.20 pts

Write down the expressions for C_l and R_l in terms of r, σ, ρ and ξ.

Answer: Capacitance C_l is the capacitance per unit length of the surface of the cylinder, so

$$C_l = 2\pi r\sigma.$$

Resistance R_l is the resistance per unit length of water around one cylinder, so $R_l = \rho/S_1$, where S_1 is the area of water per cylinder, so

$$R_l = \frac{\rho}{\frac{1}{\xi} - \pi r^2}.$$

C5 — 0.10 pts

Write the complex current $\tilde{I}(x)$ in terms of the derivative of the complex potential $\frac{d\tilde{\varphi}(x)}{dx}$ and R_l.

Let's write down the voltage drop per unit length of the conductor:

$$-R_l dx\, \tilde{I} = \tilde{\varphi}(x + dx) - \tilde{\varphi}(x)$$

Answer:

$$\tilde{I}(x) = -\frac{1}{R_l}\frac{d\tilde{\varphi}(x)}{dx}$$

C6 — 0.10 pts

Relate the derivative of the complex current $\frac{d\tilde{I}(x)}{dx}$ *and the complex potential* $\tilde{\varphi}(x)$ *via* C_l.

Current through unit length of capacitor

$$\frac{\tilde{I}(x+dx)-\tilde{I}(x)}{i\omega C_l dx} = -\tilde{\varphi}(x)$$

Answer:

$$\frac{d\tilde{I}(x)}{dx} = -i\omega C_l \tilde{\varphi}(x)$$

C7 — 0.20 pts

Show that $\tilde{I}(0)=0$.

If the current is not zero, then it flows through an infinitesimal capacitance, which means $\varphi(0)\to\infty$.

C8 — 1.25 pts

Solve the system of differential equations obtained in steps ***C5*** *and* ***C6*** *and obtain the impedance* $\tilde{Z}_c(\omega)$ *of one cylinder and the electrolyte surrounding it. Neglect the capacity of the base of the cylinder and the capacity of the flat surface on which the cylinders stand. Express your answer using* C_l, R_l, l *and* ω.

We obtain a differential equation on $\tilde{\varphi}(x)$:

$$\tilde{\varphi}'' - i\omega C_l R_l \tilde{\varphi} = 0$$

This is a homogeneous linear differential equation of the second order. Let's find our own numbers:

$$\lambda^2 = i\omega C_l R_l.$$

$$\lambda_1 = e^{\frac{i\frac{\pi}{2}}{2}}\sqrt{\omega C_l R_l} = (1+i)\sqrt{\frac{\omega R_l C_l}{2}}$$

$$\lambda_2 = e^{\frac{i\frac{\pi}{2}+2\pi i}{2}}\sqrt{\omega C_l R_l} = -(1+i)\sqrt{\frac{\omega R_l C_l}{2}}$$

Let's denote $\sqrt{\frac{\omega C_l R_l}{2}}$ as ζ. Then

$$\tilde{\varphi}(x) = Ae^{i\zeta x}e^{\zeta x} + Be^{-i\zeta x}e^{-\zeta x}.$$

Boundary conditions: $\tilde{\varphi}(l) = Ue^{i\omega t}$, $\tilde{\varphi}'(0)=0$.

$$\tilde{\varphi}'(0) = A(1+i)\zeta - B(1+i)\zeta = 0 \quad \Rightarrow \quad B = A,$$

$$\tilde{\varphi}(l) = A\left(e^{i\zeta l}e^{\zeta l} + e^{-i\zeta l}e^{-\zeta l}\right) = 2A\cosh(\zeta l + i\zeta l) = Ue^{i\omega t}$$

As a result, the solution for the potential looks like this:

$$\tilde{\varphi}(x) = Ue^{i\omega t}\frac{\cosh(\zeta x + i\zeta x)}{\cosh(\zeta l + i\zeta l)}$$

Let's find the current $\tilde{I}(l)$:

$$\tilde{I}(l) = -\frac{1}{R_l}\tilde{\varphi}'(l) = -Ue^{i\omega t}\frac{\zeta(1+i)}{R_l}\tanh(\zeta l + i\zeta l),$$

then the impedance (the minus arises due to the coordination of the positive direction of the current and the decrease in impedance):

$$\tilde{Z}_c = \frac{\tilde{\varphi}(l)}{-\tilde{I}(l)} = \frac{R_l}{\zeta(1+i)\tanh(\zeta l + i\zeta l)} = \sqrt{\frac{R_l}{2\omega C_l}}\frac{1-i}{\tanh\left((1+i)\sqrt{\frac{\omega C_l R_l}{2}}\,l\right)}$$

Let's expand tanh:

$$\tilde{Z}_c = \sqrt{\frac{R_l}{\omega C_l}} \frac{1 - i\tanh\sqrt{\frac{\omega C_l R_l}{2}}l \cdot \tan\sqrt{\frac{\omega C_l R_l}{2}}l}{\tanh\sqrt{\frac{\omega C_l R_l}{2}}l - i\tan\sqrt{\frac{\omega C_l R_l}{2}}l} e^{-i\pi/4}.$$

Answer:

$$\tilde{Z}_c = \sqrt{\frac{R_l}{\omega C_l}} \frac{1 + i\tanh\sqrt{\frac{\omega C_l R_l}{2}}l \cdot \tan\sqrt{\frac{\omega C_l R_l}{2}}l}{\tanh\sqrt{\frac{\omega C_l R_l}{2}}l + i\tan\sqrt{\frac{\omega C_l R_l}{2}}l} e^{-i\pi/4}.$$

C9 — 0.20 pts

Obtain the impedance $\tilde{Z}_c(\omega)$ to the approximation $\sqrt{\omega C_l R_l l^2} \gg 1$. Express your answer using C_l, R_l, l and ω. The following theoretical points are also solved in this approximation.

Let's multiply the numerator and denominator by $\cosh\zeta l \cdot \cos\zeta l$:

$$\tilde{Z}_c = \sqrt{\frac{R_l}{\omega C_l}} \frac{\cosh\zeta l \cos\zeta l + i\sin\zeta l \sinh\zeta l}{\cos\zeta l \sinh\zeta l + i\cosh\zeta l \sin\zeta l} e^{-i\pi/4},$$

at $\zeta l \gg 1$ the hyperbolic functions $\sinh\zeta l$ and $\cosh\zeta l$ are equal to the exponent $\frac{e^{\zeta l}}{2}$, therefore

$$\tilde{Z}_c \simeq \sqrt{\frac{R_l}{\omega C_l}} \frac{\cos\zeta l e^{\zeta l} + i\sin\zeta l e^{\zeta l}}{\cos\zeta l e^{\zeta l} + i\sin\zeta l e^{\zeta l}} e^{-i\pi/4} = \sqrt{\frac{R_l}{\omega C_l}} e^{-i\pi/4}$$

Answer:

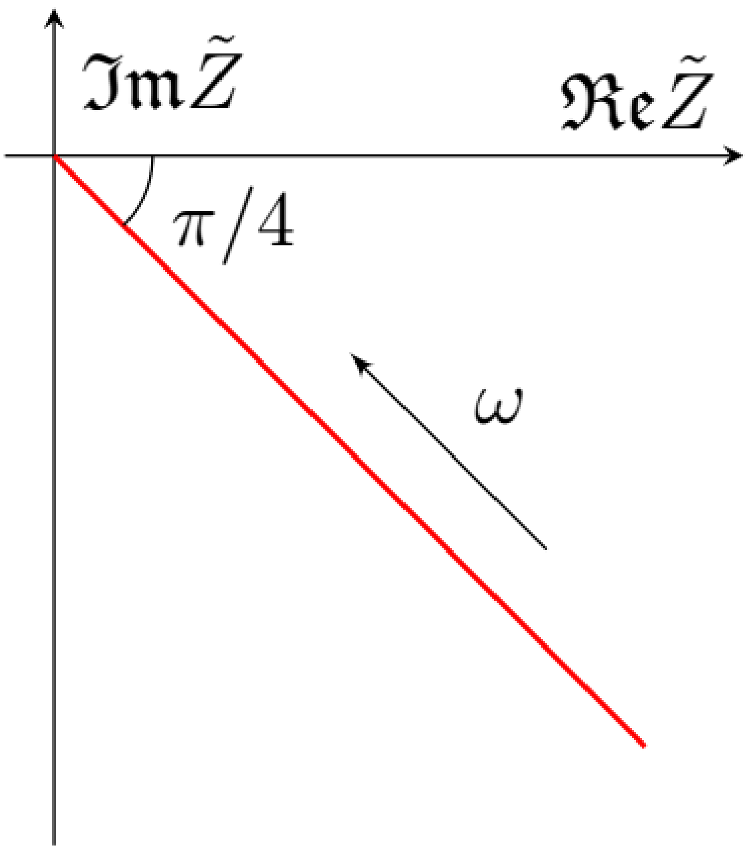
$$\tilde{Z}_c = \sqrt{\frac{R_l}{\omega C_l}} e^{-i\pi/4}$$

C10 — 0.20 pts

What is the impedance $\tilde{Z}_m(\omega)$ of one surface of the cylinder in contact with the electrolyte? Surface area S. Express your answer using r, σ, ρ, ξ, S and ω. Draw a qualitative Nyquist diagram $\tilde{Z}_m(\omega)$. The theoretical and experimental Nyquist diagrams should be identical in shape, but some of their characteristics may differ slightly.

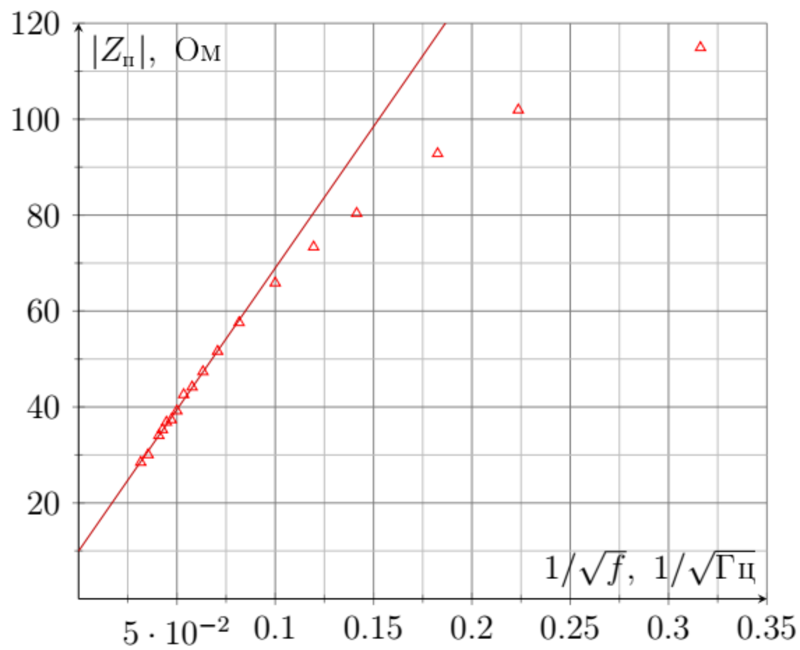
Let's combine the results of the previous paragraphs.

$$\tilde{Z}_m = \frac{1}{\xi S} \sqrt{\frac{\rho}{2\pi r \sigma \omega \left(\frac{1}{\xi} - \pi r^2\right)}} e^{-i\pi/4}$$



C11 0.70 pts

Plot a linearized graph of $|\tilde{Z}_m|$ versus ω.



C12 **0.10 pts**

Define σ.

Answer: $\sigma_0 = 6.9\,\frac{\mu\mathrm{F}}{\mathrm{cm}^2}$